

5

First Law Applied to Flow Processes

5.1 Control Volume

For any system and in any process, the first law can be written as

$$Q = \Delta E + W$$

where E represents all forms of energy stored in the system.

For a pure substance

$$E = E_K + E_P + U$$

where E_K is the K.E., E_P the P.E., and U the residual energy stored in the molecular structure of the substance.

$$Q = \Delta E_K + \Delta E_P + \Delta U + W \quad (5.1)$$

When there is mass transfer across the system boundary, the system is called an open system. Most of the engineering devices are open systems involving the flow of fluids through them.

Equation (5.1) refers to a system having a particular mass of substance, and is free to move from place to place.

Consider a steam turbine (Fig. 5.1) in which steam enters at a high pressure, does work upon the turbine rotor, and then leaves the turbine at low pressure through the exhaust pipe.

If a certain mass of steam is considered as the thermodynamic system, then the energy equation becomes

$$Q = \Delta E_K + \Delta E_P + \Delta U + W$$

and in order to analyze the expansion process in turbine the moving system is to be followed as it travels through the turbine, taking into account the work and heat interactions all the way through. This method of analysis is similar to that of Lagrange in fluid mechanics.

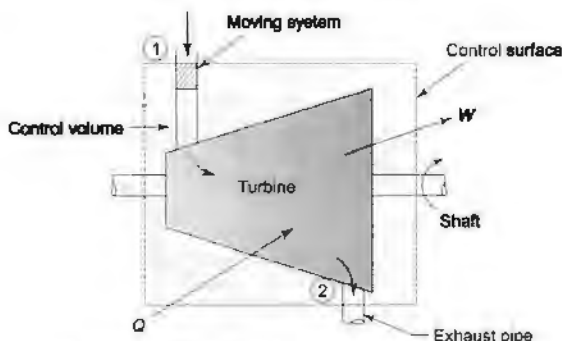


Fig. 5.1 Flow process involving work and heat interactions

Although the system approach is quite valid, there is another approach which is found to be highly convenient. Instead of concentrating attention upon a certain quantity of fluid, which constitutes a moving system in flow process, attention is focussed upon a certain fixed region in space called a *control volume* through which the moving substance flows. This is similar to the analysis of Euler in fluid mechanics.

To distinguish the two concepts, it may be noted that while the system (closed) boundary usually changes shape, position and orientation relative to the observer, the control volume boundary remains fixed and unaltered. Again, while matter usually crosses the control volume boundary, no such flow occurs across the system boundary.

The broken line in Fig. 5.1 represents the surface of the control volume which is known as the *control surface*. This is the same as the system boundary of the open system. The method of analysis is to inspect the control surface and account for all energy quantities transferred through this surface. Since there is mass transfer across the control surface, a mass balance also has to be made. Sections 1 and 2 allow mass transfer to take place, and Q and W are the heat and work interactions respectively.

5.2 Steady Flow Process

As a fluid flows through a certain control volume, its thermodynamic properties may vary along the space coordinates as well as with time. If the rates of flow of mass and energy through the control surface change with time, the mass and energy within the control volume also would change with time.

'Steady flow' means that the rates of flow of mass and energy across the control surface are constant.

In most engineering devices, there is a constant rate of flow of mass and energy through the control surface, and the control volume in course of time attains a steady state. *At the steady state of a system, any thermodynamic property will*

have a fixed value at a particular location, and will not alter with time. Thermodynamic properties may vary along space coordinates, but do not vary with time. 'Steady state' means that the state is steady or invariant with time.

5.3 Mass Balance and Energy Balance in a Simple Steady Flow Process

In Fig. 5.2, a steady flow system has been shown in which one stream of fluid enters and one stream leaves the control volume. There is no accumulation of mass or energy within the control volume, and the properties at any location within the control volume are steady with time. Sections 1.1 and 2.2 indicate, respectively, the entrance and exit of the fluid across the control surface. The following quantities are defined with reference to Fig. 5.2.

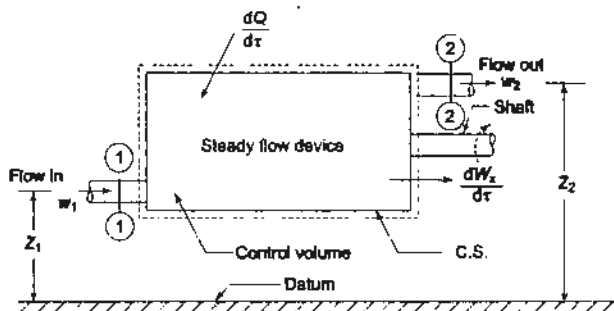


Fig. 5.2 Steady flow Process

A_1, A_2 —cross-section of stream, m^2

w_1, w_2 —mass flow rate, kg/s

p_1, p_2 —pressure, absolute, N/m^2

v_1, v_2 —specific volume, m^3/kg

u_1, u_2 —specific internal energy, J/kg

V_1, V_2 —velocity, m/s

Z_1, Z_2 —elevation above an arbitrary datum, m

$\frac{dQ}{dt}$ —net rate of heat transfer through the control surface, J/s

$\frac{dW_s}{dt}$ —net rate of work transfer through the control surface, J/s

exclusive of work done at sections 1 and 2 in transferring the fluid through the control surface.

t —time, s .

Subscripts 1 and 2 refer to the inlet and exit sections.

5.3.1 Mass Balance

By the conservation of mass, if there is no accumulation of mass within the control volume, the mass flow rate entering must equal the mass flow rate leaving, or

$$\begin{aligned} w_1 &= w_2 \\ \text{or} \quad \frac{A_1 V_1}{v_1} &= \frac{A_2 V_2}{v_2} \end{aligned} \quad (5.2)$$

This equation is known as the *equation of continuity*.

5.3.2 Energy Balance

In a flow process, the work transfer may be of two types: the *external work* and the *flow work*. The external work refers to all the work transfer across the control surface other than that due to normal fluid forces. In engineering thermodynamics the only kinds of external work of importance are *shear* (shaft or stirring) *work* and *electrical work*. In Fig. 5.2 the only external work occurs in the form of shaft work, W_x . The flow work, as discussed in Sec. 3.4, is the displacement work done by the fluid of mass dm_1 at the inlet section 1 and that of mass dm_2 at the exit section 2, which are $(-p_1 v_1 dm_1)$ and $(+p_2 v_2 dm_2)$ respectively. Therefore, the total work transfer is given by

$$W = W_x - p_1 v_1 dm_1 + p_2 v_2 dm_2 \quad (5.3)$$

In the rate form,

$$\begin{aligned} \frac{dW}{d\tau} &= \frac{dW_x}{d\tau} - p_1 v_1 \frac{dm_1}{d\tau} + p_2 v_2 \frac{dm_2}{d\tau} \\ \text{or} \quad \frac{dW}{d\tau} &= \frac{dW_x}{d\tau} - w_1 p_1 v_1 + w_2 p_2 v_2 \end{aligned} \quad (5.4)$$

Since there is no accumulation of energy, by the conservation of energy, the total rate of flow of all energy streams entering the control volume must equal the total rate of flow of all energy streams leaving the control volume. This may be expressed in the following equation.

$$w_1 e_1 + \frac{dQ}{d\tau} = w_2 e_2 + \frac{dW}{d\tau}$$

Substituting for $\frac{dW}{d\tau}$ from Eq. (5.4)

$$w_1 e_1 + \frac{dQ}{d\tau} = w_2 e_2 + \frac{dW_x}{d\tau} - w_1 p_1 v_1 + w_2 p_2 v_2$$

$$w_1 e_1 + w_1 p_1 v_1 + \frac{dQ}{d\tau} = w_2 e_2 + w_2 p_2 v_2 + \frac{dW_x}{d\tau} \quad (5.5)$$

where e_1 and e_2 refer to the energy carried into or out of the control volume with unit mass of fluid.

The specific energy e is given by

$$\begin{aligned} e &= e_k + e_p + u \\ &= \frac{V^2}{2} + Zg + u \end{aligned} \quad (5.6)$$

Substituting the expression for e in Eq. (5.5)

$$\begin{aligned} w_1 \left(\frac{V_1^2}{2} + Z_1g + u_1 \right) + w_1 p_1 v_1 + \frac{dQ}{d\tau} \\ = w_2 \left(\frac{V_2^2}{2} + Z_2g + u_2 \right) + w_2 p_2 v_2 + \frac{dW_x}{d\tau} \end{aligned}$$

$$\begin{aligned} \text{or} \quad w_1 \left(h_1 + \frac{V_1^2}{2} + Z_1g \right) + \frac{dQ}{d\tau} \\ = w_2 \left(h_2 + \frac{V_2^2}{2} + Z_2g \right) + \frac{dW_x}{d\tau} \end{aligned} \quad (5.7)$$

where

$$h = u + pv.$$

And, since

$$w_1 = w_2, \quad \text{let } w = w_1 = w_2 = \frac{dm}{d\tau}$$

Dividing Eq. (5.7) by $\frac{dm}{d\tau}$

$$\begin{aligned} h_1 + \frac{V_1^2}{2} + Z_1g + \frac{dQ}{dm} \\ = h_2 + \frac{V_2^2}{2} + Z_2g + \frac{dW_x}{dm} \end{aligned} \quad (5.8)$$

Equations (5.7) and (5.8) are known as *steady flow energy equations* (S.F.E.E.), for a single stream of fluid entering and a single stream of fluid leaving the control volume. All the terms in Eq. (5.8) represent energy flow per unit mass of fluid (J/kg) whereas all the terms in Eq. (5.7) represent energy flow per unit time (J/kg). The basis of energy flow per unit mass is usually more convenient when only a single stream of fluid enters and leaves a control volume. When more than one fluid stream is involved the basis of energy flow per unit time is more convenient.

Equation (5.8) can be written in the following form,

$$Q - W_x = (h_2 - h_1) + \frac{\bar{V}_2^2 - \bar{V}_1^2}{2} + g(Z_2 - Z_1) \quad (5.9)$$

where Q and W_x refer to energy transfer per unit mass. In the differential form, the SFEE becomes

$$\delta Q - \delta W_x = dh + \bar{V} d\bar{V} + g dZ \quad (5.10)$$

When more than one stream of fluid enters or leaves the control volume (Fig. 5.3), the mass balance and energy balance for steady flow are given below.

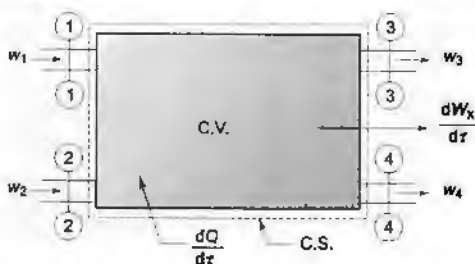


Fig. 5.3 Steady flow process involving two fluid streams at the inlet and exit of the control volume

Mass balance

$$w_1 + w_2 = w_3 + w_4 \quad (5.11)$$

$$\frac{A_1 V_1}{v_1} + \frac{A_2 V_2}{v_2} = \frac{A_3 V_3}{v_3} + \frac{A_4 V_4}{v_4} \quad (5.12)$$

Energy balance

$$\begin{aligned} w_1 \left(h_1 + \frac{V_1^2}{2} + Z_1 g \right) + w_2 \left(h_2 + \frac{V_2^2}{2} + Z_2 g \right) + \frac{\delta Q}{d\tau} \\ = w_3 \left(h_3 + \frac{V_3^2}{2} + Z_3 g \right) + w_4 \left(h_4 + \frac{V_4^2}{2} + Z_4 g \right) + \frac{\delta W_x}{d\tau} \end{aligned} \quad (5.13)$$

The steady flow energy equation applies to a wide variety of processes like pipe line flows, heat transfer processes, mechanical power generation in engines and turbines, combustion processes, and flows through nozzles and diffusers. In certain problems, some of the terms in steady flow energy equation may be negligible or zero. But it is best to write the full equation first, and then eliminate the terms which are unnecessary.

5.4 Some Examples of Steady Flow Processes

The following examples illustrate the applications of the steady flow energy equation in some of the engineering systems.

5.4.1 Nozzle and Diffusor

A nozzle is a device which increases the velocity or K.E. of a fluid at the expense of its pressure drop, whereas a diffusor increases the pressure of a fluid at the

expense of its K.E. Figure 5.4 shows a nozzle which is insulated. The steady flow energy equation of the control surface gives

$$h_1 + \frac{V_1^2}{2} + Z_1 g + \frac{dQ}{dm} = h_2 + \frac{V_2^2}{2} + Z_2 g + \frac{dW_s}{dm}$$

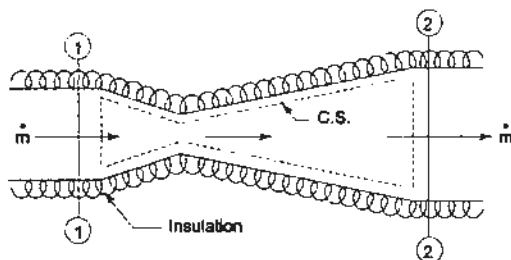


Fig. 5.4 Steady flow process involving one fluid stream at the inlet and at exit of the control volume

Here $\frac{dQ}{dm} = 0$, $\frac{dW_s}{dm} = 0$, and the change in potential energy is zero.

The equation reduces to

$$h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2} \quad (5.14)$$

The continuity equation gives

$$w = \frac{A_1 V_1}{v_1} = \frac{A_2 V_2}{v_2} \quad (5.15)$$

When the inlet velocity or the 'velocity of approach' V_1 is small compared to the exit velocity V_2 , Eq. (5.14) becomes

$$h_1 = h_2 + \frac{V_2^2}{2}$$

or

$$V_2 = \sqrt{2(h_1 - h_2)} \text{ m/s}$$

where $(h_1 - h_2)$ is in J/kg.

Equations (5.14) and (5.15) hold good for a diffuser as well.

5.4.2 Throttling Device

When a fluid flows through a constricted passage, like a partially opened valve, an orifice, or a porous plug, there is an appreciable drop in pressure, and the flow is said to be throttled. Figure 5.5 shows the process of throttling by a partially opened valve on a fluid flowing in an insulated pipe. In the steady-flow energy Eq. (5.8),

$$\frac{\delta Q}{dm} = 0, \quad \frac{\delta W_s}{dm} = 0$$

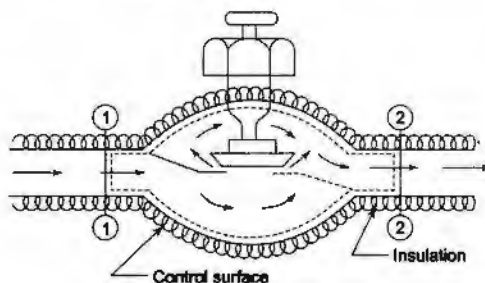


Fig. 5.5 Flow through a valve

and the changes in P.E. are very small and ignored. Thus the S.F.E.E. reduces to

$$h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2}$$

Often the pipe velocities in throttling are so low that the K.E. terms are also negligible. So

$$h_1 = h_2 \quad (5.16)$$

or the enthalpy of the fluid before throttling is equal to the enthalpy of the fluid after throttling.

5.4.3 Turbine and Compressor

Turbines and engines give positive power output, whereas compressors and pumps require power input.

For a turbine (Fig. 5.6) which is well insulated, the flow velocities are often small, and the K.E. terms can be neglected. The S.F.E.E. then becomes

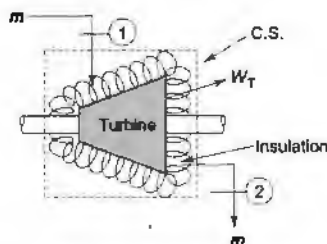


Fig. 5.6 Flow through a turbine

$$h_1 = h_2 + \frac{\delta W_x}{dm}$$

or
$$\frac{W_x}{m} = (h_1 - h_2)$$

It is seen that work is done by the fluid at the expense of its enthalpy.

Similarly, for an adiabatic pump or compressor, work is done upon the fluid and W is negative. So the S.F.E.E. becomes

$$h_1 = h_2 - \frac{W_x}{m}$$

or
$$\frac{W_x}{m} = h_2 - h_1$$

The enthalpy of the fluid increases by the amount of work input.

5.4.4 Heat Exchanger

A heat exchanger is a device in which heat is transferred from one fluid to another. Figure 5.7 shows a steam condenser, where steam condenses outside the tubes and cooling water flows through the tubes. The S.F.E.E. for the C.S. gives

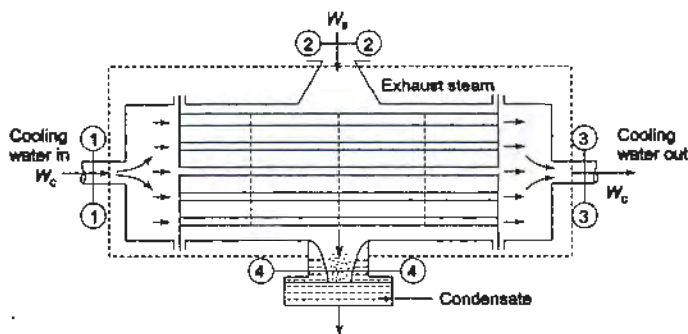


Fig. 5.7 Steam condenser

$$w_c h_1 + w_s h_2 = w_c h_3 + w_s h_4$$

or
$$w_s (h_2 - h_4) = w_c (h_3 - h_1)$$

Here the K.E. and P.E. terms are considered small, there is no external work done, and energy exchange in the form of heat is confined only between the two fluids, i.e., there is no external heat interaction or heat loss.

Figure 5.8 shows a steam desuperheater where the temperature of the superheated steam is reduced by spraying water. If w_1 , w_2 , and w_3 are the mass

flow rates of the injected water, of the steam entering, and of the steam leaving, respectively, and h_1 , h_2 , and h_3 are the corresponding enthalpies, and if K.E. and P.E. terms are neglected as before, the S.F.E.E. becomes

$$w_1 h_1 + w_2 h_2 = w_3 h_3$$

and the mass balance gives

$$w_1 + w_2 = w_3$$

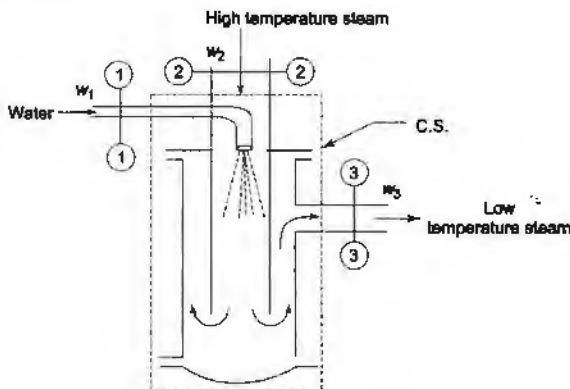


Fig. 5.8 Steam desuperheater

5.5 Comparison of S.F.E.E. with Euler and Bernoulli Equations

The steady flow energy Eq. (5.8) can be written as

$$\frac{dQ}{dm} = (h_2 - h_1) + \frac{\bar{V}_2^2 - \bar{V}_1^2}{2} + (Z_2 - Z_1)g + \frac{dW_x}{dm}$$

In the differential form the S.F.E.E. becomes

$$\delta Q = dh + \bar{V} d\bar{V} + g dZ + dW_x \quad (5.17)$$

where δQ and δW_x refer to unit mass of the substance. Since $h = u + pv$ and

$\delta Q = \delta u + p d\bar{V}$ (for a quasi-static path involving only $p d\bar{V}$ -work), Eq. (5.17) can be written as

$$du + p d\bar{V} = du + p d\bar{V} + \bar{V} dp + \bar{V} d\bar{V} + g dZ + \delta W_x$$

For an inviscid frictionless fluid flowing through a pipe

$$\bar{V} dp + \bar{V} d\bar{V} + g dZ = 0 \quad (5.18)$$

This is the *Euler equation*. If we integrate between two sections 1 and 2 of the pipe

$$\int_1^2 v dp + \int_1^2 v dV + \int_1^2 g dZ = 0$$

For an incompressible fluid, $v = \text{constant}$

$$\therefore v(p_2 - p_1) + \frac{V_2^2}{2} - \frac{V_1^2}{2} + g(Z_2 - Z_1) = 0 \quad (5.19)$$

Since the specific volume v is the reciprocal of the density ρ , we have

$$\frac{p_1}{\rho} + \frac{V_1^2}{2} + Z_1 g = \frac{p_2}{\rho} + \frac{V_2^2}{2} + Z_2 g \quad (5.20)$$

$$\text{or} \quad \frac{p}{\rho} + \frac{V^2}{2} + Zg = \text{constant} \quad (5.21)$$

This is known as the *Bernoulli equation*, which is valid for an inviscid incompressible fluid. It can also be expressed in the following form

$$\Delta \left(p v + \frac{V^2}{2} + gZ \right) = 0 \quad (5.22)$$

where v is constant and $\Delta(\dots)$ means 'increase in...

The S.F.E.E. as given by Eq. (5.18) or Eq. (5.17) can be written with $(u + pv)$ substituted for h , as follows:

$$Q - W_s = \Delta \left(u + pv + \frac{V^2}{2} + gZ \right) \quad (5.23)$$

A comparison of Eqs (5.22) and (5.23) shows that they have several terms in common. However, while the Bernoulli equation is restricted to frictionless incompressible fluids, the S.F.E.E. is not, and is valid for viscous compressible fluids as well. The Bernoulli equation is, therefore, a special limiting case of the more general steady flow energy equation.

5.6 Variable Flow Processes

Many flow processes, such as filling up and evacuating gas cylinders, are not steady. Such processes can be analyzed by the control volume technique. Consider a device through which a fluid is flowing under non-steady state conditions (Fig. 5.9). The rate at which the mass of fluid within the control volume is accumulated as equal to the net rate of mass flow across the control surface, as given below

$$\frac{dm_v}{d\tau} = w_1 - w_2 = \frac{dm_1}{d\tau} - \frac{dm_2}{d\tau} \quad (5.24)$$

where m_v is the mass of fluid within the control volume at any instant.

Over any finite period of time

$$\Delta m_V = \Delta m_1 - \Delta m_2 \quad (5.25)$$

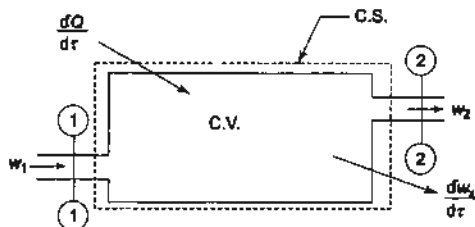


Fig. 5.9 Variable flow process

The rate of accumulation of energy within the control volume is equal to the net rate of energy flow across the control surface. If E_V is the energy of fluid within the control volume at any instant.

Rate of energy increase = Rate of energy inflow – Rate of energy outflow

$$\begin{aligned} \frac{dE_V}{d\tau} &= w_1 \left(h_1 + \frac{V_1^2}{2} + Z_1 g \right) + \frac{dQ}{d\tau} \\ &\quad - w_2 \left(h_2 + \frac{V_2^2}{2} + Z_2 g \right) - \frac{dW_x}{d\tau} \end{aligned} \quad (5.26)$$

Now

$$E_V = \left(U + \frac{m V^2}{2} + m g Z \right)_V$$

where m is the mass of fluid in the control volume at any instant.

$$\begin{aligned} \therefore \frac{dE_V}{d\tau} &= \frac{d}{d\tau} \left(U + \frac{m V^2}{2} + m g Z \right)_V \\ &= \left(h_1 + \frac{V_1^2}{2} + Z_1 g \right) \frac{dm_1}{d\tau} + \frac{dQ}{d\tau} \\ &\quad - \left(h_2 + \frac{V_2^2}{2} + Z_2 g \right) \frac{dm_2}{d\tau} - \frac{dW_x}{d\tau} \end{aligned} \quad (5.27)$$

Figure 5.10 shows all these energy flux quantities. For any finite time interval, equation (5.27) becomes

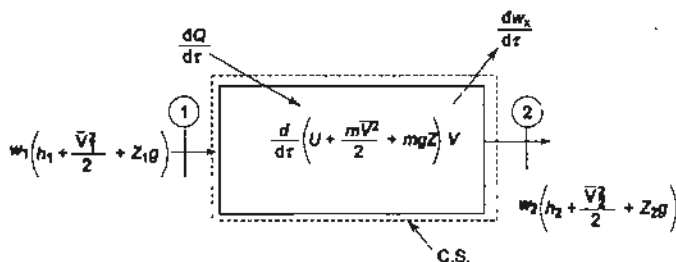


Fig. 5.10 Energy fluxes in an unsteady system

$$\Delta E_V = Q - W_x + \int \left(h_1 + \frac{V_1^2}{2} + Z_1 g \right) dm_1 - \int \left(h_2 + \frac{V_2^2}{2} + Z_2 g \right) dm_2 \quad (5.28)$$

Equation (5.26) is the *general energy equation*. For steady flow,

$$\frac{dE_V}{d\tau} = 0,$$

and the equation reduces to Eq. (5.7). For a closed system $w_1 = 0$, $w_2 = 0$, then from Eq. (5.26),

$$\frac{dE_V}{d\tau} = \frac{dQ}{d\tau} - \frac{dW_x}{d\tau}$$

or

$$dE_V = dQ - dW_x \text{ or, } dQ = dE + dW_x$$

as obtained earlier.

5.7 Example of a Variable Flow Problem

Variable flow processes may be analyzed either by the system technique or the control volume technique, as illustrated below.

Consider a process in which a gas bottle is filled from a pipeline (Fig. 5.11). In the beginning the bottle contains gas of mass m_1 at state p_1 , t_1 , v_1 , h_1 and u_1 . The valve is opened and gas flows into the bottle till the mass of gas in the bottle is m_2 at state p_2 , t_2 , v_2 , h_2 and u_2 . The supply to the pipeline is very large so that the state of gas in the pipeline is constant at p_p , t_p , v_p , h_p , u_p and V_p .

System Technique Assume an envelope (which is extensible) of gas in the pipeline and the tube which would eventually enter the bottle, as shown in Fig. 5.11.

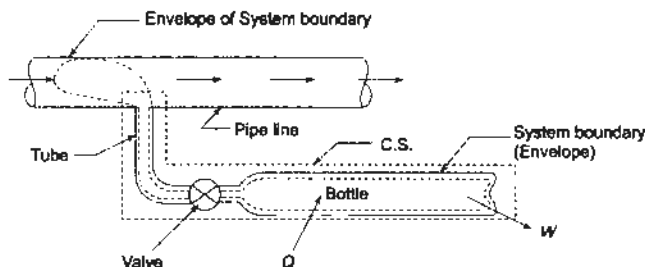


Fig. 5.11 Bottle-filling process

Energy of the gas before filling

$$E_1 = m_1 u_1 + (m_2 - m_1) \left(\frac{V_p^2}{2} + u_p \right)$$

where $(m_2 - m_1)$ is the mass of gas in the pipeline and tube which would enter the bottle.

Energy of the gas after filling

$$E_2 = m_2 u_2$$

$$\Delta E = E_2 - E_1 = m_2 u_2 - \left[m_1 u_1 + (m_2 - m_1) \left(\frac{V_p^2}{2} + u_p \right) \right] \quad (5.29)$$

The P.E. terms are neglected: The gas in the bottle is not in motion, and so the K.E. terms have been omitted.

Now, there is a change in the volume of gas because of the collapse of the envelope to zero volume. Then the work done

$$\begin{aligned} W &= p_p (V_2 - V_1) = p_p [0 - (m_2 - m_1) v_p] \\ &= -(m_2 - m_1) p_p v_p \end{aligned}$$

$\therefore$ Using the first law for the process

$$\begin{aligned} Q &= \Delta E + W \\ &= m_2 u_2 - m_1 u_1 - (m_2 - m_1) \left(\frac{V_p^2}{2} + u_p \right) - (m_2 - m_1) p_p v_p \\ &= m_2 u_2 - m_1 u_1 - (m_2 - m_1) \left(\frac{V_p^2}{2} + h_p \right) \end{aligned} \quad (5.30)$$

which gives the energy balance for the process.

Control Volume Technique Assume a control volume bounded by a control surface, as shown in Fig. 5.11. Applying the energy Eq. (5.27) to this case, the following energy balance may be written on a time rate basis

$$\frac{dE_v}{d\tau} = \frac{dQ}{d\tau} + \left(h_p + \frac{V_p^2}{2} \right) \frac{dm}{d\tau}$$

Since h_p and V_p are constant, the equation is integrated to give for the total process

$$\Delta E_v = Q + \left(h_p + \frac{V_p^2}{2} \right) (m_2 - m_1)$$

Now

$$\Delta E_v = U_2 - U_1 = m_2 u_2 - m_1 u_1$$

$\therefore$

$$Q = m_2 u_2 - m_1 u_1 - \left(h_p + \frac{V_p^2}{2} \right) (m_2 - m_1)$$

This equation is the same as Eq. (5.30).

If $m_1 = 0$, i.e., the bottle is initially evacuated,

$$Q = m_2 u_2 - m_2 \left(h_p + \frac{V_p^2}{2} \right)$$

Again, if $Q = 0$ and $h_p \gg \frac{V_p^2}{2}$,

$$0 = m_2 u_2 - m_2 h_p$$

or,

$$u_2 = h_p = u_p + p_p v_p$$

Thus, flow work ($p_p v_p$) is converted to increase in molecular internal energy ($u_2 - u_p$).

If the gas is assumed ideal,

$$c_v T_2 = c_p T_p$$

or,

$$T_2 = \gamma T_p$$

If $T_p = 27 + 273 = 300$ K, then for air

$$T_2 = 1.4 \times 300 = 420 \text{ K}$$

or,

$$t_2 = 147^\circ\text{C}$$

Therefore, in adiabatically filling a bottle with air at 27°C , the gas temperature rises to 147°C due to the flow work being converted to internal energy increase.

5.8 Discharging and Charging a Tank

Let us consider a tank discharging a fluid into a supply line (Fig. 5.12). Since

$dW_x = 0$ and $dm_{in} = 0$, applying first law to the control volume,

$$dU_v = dQ + \left(h + \frac{V^2}{2} + gz \right)_{out} dm_{out} \quad (5.31)$$

Assuming K.E. and P.E. of the fluid to be small and $dQ = 0$

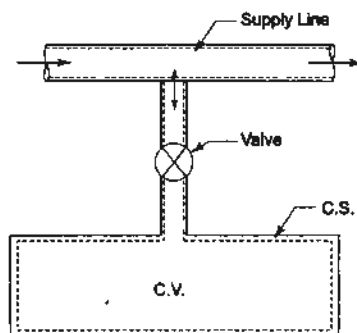


Fig. 5.12 Charging and discharging a tank

$$d(mu) = hdm$$

$$mdu + udm = udm + pv dm$$

$$\frac{dm}{m} = \frac{du}{pv} \quad (5.32)$$

Again

$$V = vm = \text{const.}$$

$$vdm + m dv = 0$$

or

$$\frac{dm}{m} = - \frac{dv}{v} \quad (5.33)$$

From Eqs (5.32) and (5.33),

$$\frac{du}{pv} = - \frac{dv}{v}$$

$$d(u + pv) = 0$$

or

$$\delta Q = 0$$

which shows that the process is adiabatic and quasi-static.

For charging the tank

$$\int (h dm)_m = \Delta U_V = m_2 u_2 - m_1 u_1 \quad (5.34)$$

$$m_p h_p = m_2 u_2 - m_1 u_1$$

where the subscript p refers to the constant state of the fluid in the pipeline. If the tank is initially empty, $m_1 = 0$.

$$m_p h_p = m_2 u_2$$

Since

$$m_p = m_2$$

$$h_p = u_2$$

Enthalpy is converted to internal energy.

If the fluid is an ideal gas, the temperature of the gas in the tank after it is charged is given by

$$\begin{aligned} c_p T_p &= c_v T_2 \\ \text{or} \quad T_2 &= \gamma T_p \end{aligned} \quad (5.35)$$

SOLVED EXAMPLES

Example 5.1 Air flows steadily at the rate of 0.5 kg/s through an air compressor, entering at 7 m/s velocity, 100 kPa pressure, and 0.95 m³/kg volume, and leaving at 5 m/s, 700 kPa, and 0.19 m³/kg. The internal energy of the air leaving is 90 kJ/kg greater than that of the air entering. Cooling water in the compressor jackets absorbs heat from the air at the rate of 58 kW. (a) Compute the rate of shaft work input to the air in kW. (b) Find the ratio of the inlet pipe diameter to outlet pipe diameter.

Solution Figure Ex. 5.1 shows the details of the problem.

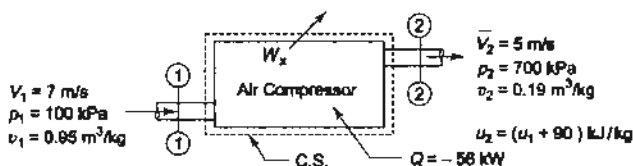


Fig. Ex. 5.1

(a) Writing the steady flow energy equation, we have

$$\begin{aligned} w \left(u_1 + p_1 v_1 + \frac{V_1^2}{2} + Z_1 g \right) + \frac{dQ}{d\tau} \\ = w \left(u_2 + p_2 v_2 + \frac{V_2^2}{2} + Z_2 g \right) + \frac{dW_x}{d\tau} \\ \therefore \frac{dW_x}{d\tau} = -w \left[(u_2 - u_1) + (p_2 v_2 - p_1 v_1) + \frac{V_2^2 - V_1^2}{2} + (Z_2 - Z_1)g \right] + \frac{dQ}{d\tau} \\ \therefore \frac{dW_x}{d\tau} = -0.5 \frac{\text{kg}}{\text{s}} \left[90 \frac{\text{kJ}}{\text{kg}} + (7 \times 0.19 - 1 \times 0.95) 100 \frac{\text{kJ}}{\text{kg}} \right. \\ \left. + \frac{(5^2 - 7^2) \times 10^{-3}}{2} \frac{\text{kJ}}{\text{kg}} + 0 \right] - 58 \text{ kW} \\ = -0.5 [90 + 38 - 0.012] \text{ kJ/s} - 58 \text{ kW} \\ = -122 \text{ kW} \end{aligned}$$

Ans. (a)

Rate of work input is 122 kW.

(b) From mass balance, we have

$$w = \frac{A_1 V_1}{v_1} = \frac{A_2 V_2}{v_2}$$

$$\therefore \frac{A_1}{A_2} = \frac{v_1}{v_2} \cdot \frac{V_2}{V_1} = \frac{0.95}{0.19} \times \frac{5}{7} = 3.57$$

$$\therefore \frac{d_1}{d_2} = \sqrt{3.57} = 1.89 \quad \text{Ans. (b)}$$

Example 5.2 In a steady flow apparatus, 135 kJ of work is done by each kg of fluid. The specific volume of the fluid, pressure, and velocity at the inlet are $0.37 \text{ m}^3/\text{kg}$, 600 kPa, and 16 m/s. The inlet is 32 m above the floor, and the discharge pipe is at floor level. The discharge conditions are $0.62 \text{ m}^3/\text{kg}$, 100 kPa, and 270 m/s. The total heat loss between the inlet and discharge is 9 kJ/kg of fluid. In flowing through this apparatus, does the specific internal energy increase or decrease, and by how much?

Solution Writing the steady flow energy equation for the control volume, as shown in Fig. Ex. 5.2.

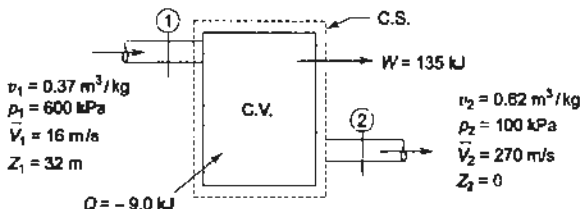


Fig. Ex. 5.2

$$u_1 + p_1 v_1 + \frac{V_1^2}{2} + Z_1 g + \frac{dQ}{dm} = u_2 + p_2 v_2 + \frac{V_2^2}{2} + Z_2 g + \frac{dW_s}{dm}$$

$$\therefore u_1 - u_2 = (p_2 v_2 - p_1 v_1) + \frac{V_2^2 - V_1^2}{2} + (Z_2 - Z_1)g + \frac{dW_s}{dm} - \frac{dQ}{dm}$$

$$= (1 \times 0.62 - 6 \times 0.37) \times 10^2 + \frac{(270^2 - 16^2) \times 10^{-3}}{2}$$

$$+ (-32 \times 9.81 \times 10^{-3}) + 135 - (-9.0)$$

$$= -160 + 36.45 - 0.314 + 135 + 9$$

$$= 20.136 \text{ kJ/kg}$$

Specific internal energy decreases by 20.136 kJ.

Example 5.3 In a steam power station, steam flows steadily through a 0.2 m diameter pipeline from the boiler to the turbine. At the boiler end, the steam conditions are found to be: $p = 4 \text{ MPa}$, $t = 400^\circ\text{C}$, $h = 3213.6 \text{ kJ/kg}$, and $v = 0.073 \text{ m}^3/\text{kg}$. At the turbine end, the conditions are found to be: $p = 3.5 \text{ MPa}$, $t = 392^\circ\text{C}$, $h = 3202.6 \text{ kJ/kg}$, and $v = 0.084 \text{ m}^3/\text{kg}$. There is a heat loss of 8.5 kJ/kg from the pipeline. Calculate the steam flow rate.

Solution Writing the steady flow energy equation for the control volume as shown in Fig. Ex. 5.3

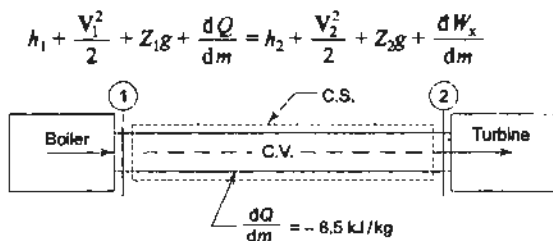


Fig. Ex. 5.3

Here, there is no change in datum, so change in potential energy will be zero.

$$\text{Now} \quad \frac{A_1 V_1}{v_1} = \frac{A_2 V_2}{v_2}$$

$$\therefore \quad V_2 = \frac{A_1 V_1}{v_1} \cdot \frac{v_2}{A_2} = \frac{v_2}{v_1} \cdot V_1 = \frac{0.084}{0.073} V_1 = 1.15 V_1$$

$$\text{and} \quad \frac{dW_x}{dm} = 0$$

$$h_1 + \frac{V_1^2}{2} + \frac{dQ}{dm} = h_2 + \frac{V_2^2}{2}$$

$$\therefore \quad \frac{(V_2^2 - V_1^2) \times 10^{-3}}{2} = h_1 - h_2 + \frac{dQ}{dm}$$

$$= 3213.6 - 3202.6 + (-8.5)$$

$$= 2.5 \text{ kJ/kg}$$

$$V_1^2 (1.15^2 - 1^2) = 5 \times 10^3$$

$$V_1^2 = 15,650 \text{ m}^2/\text{s}^2$$

$$\therefore \quad V_1 = 125.1 \text{ m/s}$$

$$\therefore \text{ Mass flow rate } w = \frac{A_1 V_1}{v_1} = \frac{\frac{\pi}{4} \times (0.2)^2 \text{ m}^2 \times 125.1 \text{ m/s}}{0.073 \text{ m}^3/\text{kg}}$$

$$= 53.8 \text{ kg/s}$$

Ans.

Example 5.4 A certain water heater operates under steady flow conditions receiving 4.2 kg/s of water at 75°C temperature, enthalpy 313.93 kJ/kg. The water is heated by mixing with steam which is supplied to the heater at temperature 100.2°C and enthalpy 2676 kJ/kg. The mixture leaves the heater as liquid water at temperature 100°C and enthalpy 419 kJ/kg. How much steam must be supplied to the heater per hour?

Solution By mass balance across the control surface (Fig. Ex. 5.4)

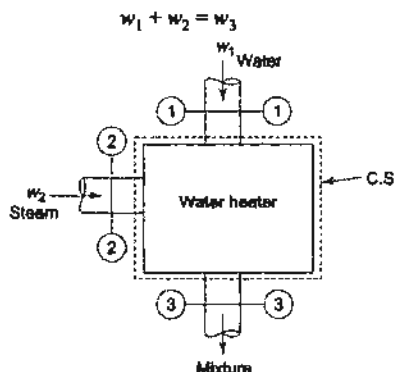


Fig. Ex. 5.4

By energy balance

$$w_1 \left(h_1 + \frac{V_1^2}{2} + Z_1 g \right) + \frac{dQ}{d\tau} + w_2 \left(h_2 + \frac{V_2^2}{2} + Z_2 g \right) \\ = w_3 \left(h_3 + \frac{V_3^2}{2} + Z_3 g \right) + \frac{dW_s}{d\tau}$$

By the nature of the process, there is no shaft work. Potential and kinetic energy terms are assumed to balance zero. The heater is assumed to be insulated. So the steady flow energy equation reduces to

$$w_1 h_1 + w_2 h_2 = w_3 h_3$$

$$4.2 \times 313.93 + w_2 \times 2676 = (4.2 + w_2) 419$$

$$\therefore w_2 = 0.196 \text{ kg/s}$$

$$= 705 \text{ kg/h}$$

Ans.

Example 5.5 Air at a temperature of 15°C passes through a heat exchanger at a velocity of 30 m/s where its temperature is raised to 800°C. It then enters a turbine with the same velocity of 30 m/s and expands until the temperature falls to 650°C. On leaving the turbine, the air is taken at a velocity of 60 m/s to a nozzle

where it expands until the temperature has fallen to 500°C. If the air flow rate is 2 kg/s, calculate (a) the rate of heat transfer to the air in the heat exchanger, (b) the power output from the turbine assuming no heat loss, and (c) the velocity at exit from the nozzle, assuming no heat loss. Take the enthalpy of air as $h = c_p t$, where c_p is the specific heat equal to 1.005 kJ/kg K and t the temperature.

Solution As shown in Fig. Ex. 5.5, writing the S.F.E.E. for the heat exchanger and eliminating the terms not relevant,

$$w \left(h_1 + \frac{V_1^2}{2} + Z_1 g \right) + Q_{1-2} = w \left(h_2 + \frac{V_2^2}{2} + Z_2 g \right) + W_{1-2}$$

$$\therefore wh_1 + Q_{1-2} = wh_2$$

$$\begin{aligned} Q_{1-2} &= w(h_2 - h_1) = w c_p (t_2 - t_1) \\ &= 2 \times 1.005 (800 - 15) \\ &= 2.01 \times 785 \\ &= 1580 \text{ kJ/s} \end{aligned}$$

Ans. (a).

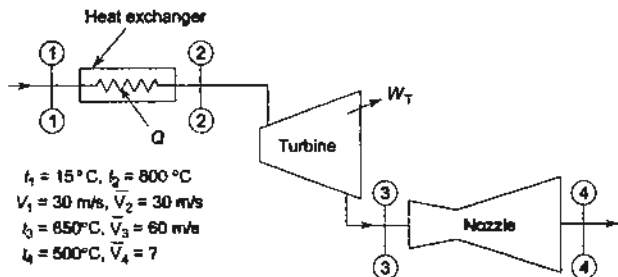


Fig. Ex. 5.5

Energy equation for the turbine gives

$$w \left(\frac{V_2^2}{2} + h_2 \right) = wh_3 + w \frac{V_3^2}{2} + W_T$$

$$\frac{V_2^2 - V_3^2}{2} + (h_2 - h_3) = W_T/w$$

$$\frac{(30^2 - 60^2) \times 10^{-3}}{2} + 1.005 (800 - 650) = W_T/w$$

$$\therefore \frac{W_T}{w} = -1.35 + 150.75$$

$$= 149.4 \text{ kJ/kg}$$

$$\begin{aligned} \therefore W_T &= 149.4 \times 2 \text{ kJ/s} \\ &= 298.8 \text{ kW} \end{aligned}$$

Ans. (b)

Writing the energy equation for the nozzle

$$\begin{aligned}\frac{V_3^2}{2} + h_3 &= \frac{V_4^2}{2} + h_4 \\ \frac{V_4^2 - V_3^2}{2} &= c_p(t_3 - t_4) \\ V_4^2 - V_3^2 &= 1.005 (650 - 500) \times 2 \times 10^3 \\ &= 301.50 \times 10^3 \text{ m}^2/\text{s}^2 \\ V_4^2 &= 30.15 \times 10^4 + 0.36 \times 10^4 \\ &= 30.51 \times 10^4 \text{ m}^2/\text{s}^2\end{aligned}$$

∴ Velocity at exit from the nozzle

$$V_4 = 554 \text{ m/s}$$

Ans. (c)

Example 5.6 The air speed of a turbojet engine in flight is 270 m/s. Ambient air temperature is -15°C . Gas temperature at outlet of nozzle is 600°C . Corresponding enthalpy values for air and gas are respectively 260 and 912 kJ/kg. Fuel-air ratio is 0.0190. Chemical energy of the fuel is 44.5 MJ/kg. Owing to incomplete combustion 5% of the chemical energy is not released in the reaction. Heat loss from the engine is 21 kJ/kg of air. Calculate the velocity of the exhaust jet.

Solution Energy equation for the turbojet engine (Fig. Ex. 5.6) gives

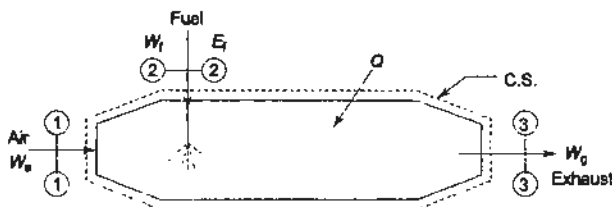


Fig. Ex. 5.6

$$\begin{aligned}w_a \left(h_a + \frac{V_a^2}{2} \right) + w_f E_f + Q &= w_g \left(h_g + \frac{V_g^2}{2} + E_g \right) \\ \left(260 + \frac{270^2 \times 10^{-3}}{2} \right) + 0.0190 \times 44500 - 21 \\ &= 1.0190 \left(912 + \frac{V_g^2 \times 10^{-3}}{2} + 0.05 \frac{0.019}{1.019} \times 44500 \right) \\ 260 + 36.5 + 845 - 21 &= 1.019 \left(912 + \frac{V_g^2 \times 10^{-3}}{2} + 42 \right)\end{aligned}$$

$$\therefore \frac{V_g^2}{2} = 156 \times 10^3 \text{ m}^2/\text{s}^2$$

$$V_g = \sqrt{31.2} \times 100 \text{ m/s}$$

Velocity of exhaust gas, $V_g = 560 \text{ m/s}$

Ans.

Example 5.7 In a reciprocating engine, the mass of gas occupying the clearance volume is m_c kg at state p_1, u_1, v_1 and h_1 . By opening the inlet valve, m_f kg of gas is taken into the cylinder, and at the conclusion of the intake process the state of the gas is given by p_2, u_2, v_2, h_2 . The state of the gas in the supply pipe is constant and is given by p_p, u_p, v_p, h_p, V_p . How much heat is transferred between the gas and the cylinder walls during the intake process?

Solution Let us consider the control volume as shown in Fig. Ex. 5.7. Writing the energy balance on a time rate basis

$$\frac{dE_v}{d\tau} = \frac{dQ}{d\tau} - \frac{dW}{d\tau} + \left(h_p + \frac{V_p^2}{2} \right) \frac{dm}{d\tau}$$

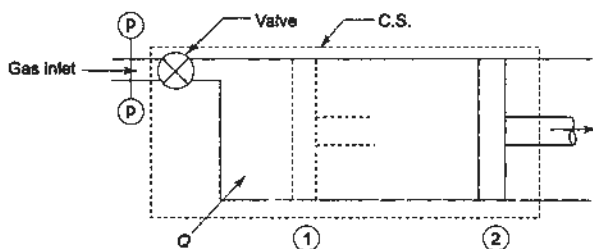


Fig. Ex. 5.7

With h_p and V_p being constant, the above equation can be integrated to give for the total process

$$\Delta E_v = Q - W + \left(h_p + \frac{V_p^2}{2} \right) m_f$$

Now

$$\Delta E_v = U_2 - U_1 = (m_c + m_f) u_2 - m_c u_1$$

$$\therefore Q = (m_c + m_f) u_2 - m_c u_1 - m_f \left(h_p + \frac{V_p^2}{2} \right) + W \quad \text{Ans.}$$

Example 5.8 The internal energy of air is given by

$$u = u_0 + 0.718t$$

where u is in kJ/kg, u_0 is any arbitrary value of u at 0°C , kJ/kg, and t is the temperature in $^\circ\text{C}$. Also for air, $p_v = 0.287 (t + 273)$, where p is in kPa and v is in m^3/kg .

A mass of air is stirred by a paddle wheel in an insulated constant volume tank. The velocities due to stirring make a negligible contribution to the internal energy of the air. Air flows out through a small valve in the tank at a rate controlled to keep the temperature in the tank constant. At a certain instant the conditions are as follows: tank volume 0.12 m^3 , pressure 1 MPa, temperature 150°C , and power to paddle wheel 0.1 kW . Find the rate of flow of air out of the tank at this instant.

Solution Writing the energy balance for the control volume as shown in Fig. Ex. 5.8

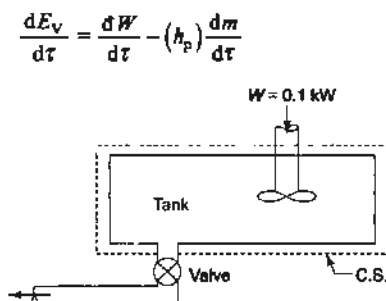


Fig. Ex. 5.8

Since there is no change in internal energy of air in the tank,

$$h_p \frac{dm}{d\tau} = \frac{\delta W}{d\tau}$$

where

$$h_p = u + pv.$$

Let

$$u = 0 \text{ at } t = 0 \text{ K} = -273^\circ\text{C}$$

$$u = u_0 + 0.718 t$$

$$0 = u_0 + 0.718 (-273)$$

$$u_0 = 0.718 \times 273 \text{ kJ/kg}$$

At $^\circ\text{C}$

$$u = 0.718 \times 273 + 0.718 t$$

$$= 0.718 \times (t + 273) \text{ kJ/kg}$$

$$h_p = 0.718 (t + 273) + 0.287 (t + 273)$$

or

$$h_p = 1.005 (t + 273)$$

At 150°C

$$h_p = 1.005 \times 423$$

$$= 425 \text{ kJ/kg}$$

$\therefore$

$$\frac{dm}{d\tau} = \frac{1}{h_p} \frac{\delta W}{d\tau}$$

$$\begin{aligned}
 &= \frac{0.1 \text{ kJ/s}}{425 \text{ kJ/kg}} = 0.236 \times 10^{-3} \text{ kg/s} \\
 &= 0.845 \text{ kg/h}
 \end{aligned}$$

This is the rate at which air flows out of the tank.

Example 5.9 A well-insulated vessel of volume V contains a gas at pressure p_0 and temperature t_0 . The gas from a main at a uniform temperature t_1 is pumped into the vessel and the inflow rate decreases exponentially with time according to $\dot{m} = \dot{m}_0 e^{-a\tau}$, where a is a constant. Determine the pressure and temperature of the gas in the vessel as a function of time. Neglect the K.E. of the gas entering the vessel and assume that the gas follows the relation

$$pv = RT, \text{ where } T = t + 273$$

and its specific heats are constant.

- If the vessel was initially evacuated, show that the temperature inside the vessel is independent of time.
- Determine the charging time when the pressure inside the vessel reaches that of the main.

Solution Since the vessel is well-insulated, $\dot{Q} = 0$ and there is no external work transfer, $\dot{W} = 0$. Therefore,

$$\frac{dE_v}{d\tau} = h_1 \frac{dm}{d\tau} = h_1 \dot{m}_0 e^{-a\tau}$$

where h_1 is the enthalpy of the gas in the main.

On integration,

$$E = E_0 + \frac{h_1 \dot{m}_0}{a} (1 - e^{-a\tau})$$

where E_0 is the initial energy of the vessel at the beginning of the charging process, i.e. $E = E_0$ at $\tau = 0$. Neglecting K.E. and P.E. changes, by energy balance

$$Mu = M_0 u_0 + \frac{\dot{m}_0}{a} (1 - e^{-a\tau}) (u_1 + p_1 v_1) \quad (1)$$

Again,

$$\frac{dm}{d\tau} = \dot{m}_0 e^{-a\tau}$$

On integration,

$$M = M_0 + \frac{\dot{m}_0 (1 - e^{-a\tau})}{a} \quad (2)$$

where M_0 is the initial mass of the gas. Eliminating M from Eqs (1) and (2),

$$\begin{aligned}
 &\left\{ M_0 + \frac{\dot{m}_0}{a} (1 - e^{-a\tau}) \right\} u - M_0 u_0 \\
 &= \frac{\dot{m}_0}{a} (1 - e^{-a\tau}) (u_1 + RT_1)
 \end{aligned}$$

$$M_0 c_v (T - T_0) = \frac{\dot{m}_0}{a} (1 - e^{-a\tau}) \{c_v(T_1 - T) + RT_1\}$$

$$T = \frac{M_0 c_v T_0 + \frac{\dot{m}_0}{a} (1 - e^{-a\tau}) c_p T_1}{\left\{ M_0 + \frac{\dot{m}_0}{a} (1 - e^{-a\tau}) \right\} c_v}$$

$$p = \frac{MRT}{V} = \frac{R}{V c_v} \left\{ M_0 c_v T_0 + \frac{\dot{m}_0}{a} (1 - e^{-a\tau}) c_p T_1 \right\}$$

$$= p_0 + \frac{\dot{m}_0 R}{a V} (1 - e^{-a\tau}) \gamma T_1$$

The above two equations show the temperature and pressure of the gas in the vessel as functions of time.

- (i) If $M_0 = 0$, $T = \gamma T_1$, i.e., the temperature inside the vessel becomes independent of time and is equal to γT_1 throughout the charging process.
- (ii) The charging process will stop when pressure inside the vessel reaches that of the main. The charging time can be found by setting $p = p_1$ in the pressure relation

$$p_1 - p_0 = \frac{\dot{m}_0 R \gamma T_1}{a V} - \frac{\dot{m}_0 R}{a V} e^{-a\tau} \gamma T_1$$

By rearrangement,

$$e^{a\tau} = \frac{\dot{m}_0 R \gamma T_1 / (a V)}{\left(\frac{\dot{m}_0 R \gamma T_1}{a V} \right) - (p_1 - p_0)}$$

$$\tau = -\frac{1}{a} \ln \left[1 - (p_1 - p_0) \frac{a V}{\dot{m}_0 R \gamma T_1} \right]$$

REVIEW QUESTIONS

- 5.1 Explain the system approach and the control volume approach in the analysis of a flow process.
- 5.2 What is a steady flow process? What is steady state?
- 5.3 Write the steady flow energy equation for a single stream entering and a single stream leaving a control volume and explain the various terms in it.
- 5.4 Give the differential form of the S.F.E.E.
- 5.5 Under what conditions does the S.F.E.E. reduce to Euler's equation?
- 5.6 How does Bernoulli's equation compare with S.F.E.E.?
- 5.7 What will be the velocity of a fluid leaving a nozzle, if the velocity of approach is very small?
- 5.8 Show that the enthalpy of a fluid before throttling is equal to that after throttling.
- 5.9 Write the general energy equation for a variable flow process.
- 5.10 What is the system technique in a bottle-filling process?
- 5.11 Explain the control volume technique in a variable flow process.

PROBLEMS

- 5.1 A blower handles 1 kg/s of air at 20°C and consumes a power of 15 kW. The inlet and outlet velocities of air are 100 m/s and 150 m/s respectively. Find the exit air temperature, assuming adiabatic conditions. Take c_p of air as 1.005 kJ/kg·K.

Ans. 28.38°C

- 5.2 A turbine operates under steady flow conditions, receiving steam at the following state: pressure 1.2 MPa, temperature 188°C, enthalpy 2785 kJ/kg, velocity 33.3 m/s and elevation 3 m. The steam leaves the turbine at the following state: pressure 20 kPa, enthalpy 2512 kJ/kg, velocity 100 m/s, and elevation 0 m. Heat is lost to the surroundings at the rate of 0.29 kJ/s. If the rate of steam flow through the turbine is 0.42 kg/s, what is the power output of the turbine in kW?

Ans. 112.51 kW

- 5.3 A nozzle is a device for increasing the velocity of a steadily flowing stream. At the inlet to a certain nozzle, the enthalpy of the fluid passing is 3000 kJ/kg and the velocity is 60 m/s. At the discharge end, the enthalpy is 2762 kJ/kg. The nozzle is horizontal and there is negligible heat loss from it. (a) Find the velocity at exit from the nozzle. (b) If the inlet area is 0.1 m² and the specific volume at inlet is 0.187 m³/kg, find the mass flow rate. (c) If the specific volume at the nozzle exit is 0.498 m³/kg, find the exit area of the nozzle.

Ans. (a) 692.5 m/s, (b) 32.08 kg/s (c) 0.023 m²

- 5.4 In an oil cooler, oil flows steadily through a bundle of metal tubes submerged in a steady stream of cooling water. Under steady flow conditions, the oil enters at 90°C and leaves at 30°C, while the water enters at 25°C and leaves at 70°C. The enthalpy of oil at t°C is given by

$$h = 1.68 t + 10.5 \times 10^{-4} t^2 \text{ kJ/kg}$$

What is the cooling water flow required for cooling 2.78 kg/s of oil?

Ans. 1.473 kg/s

- 5.5 A thermoelectric generator consists of a series of semiconductor elements (Fig. P. 5.5), heated on one side and cooled on the other. Electric current flow is produced as a result of energy transfer as heat. In a particular experiment the

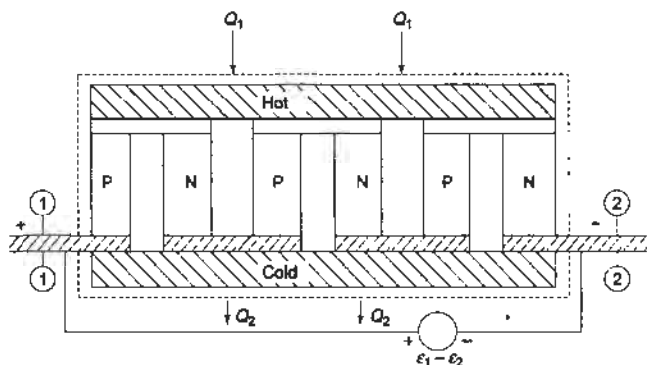


Fig. P.5.5

current was measured to be 0.5 amp and the electrostatic potential at (1) was 0.8 volt above that at (2). Energy transfer as heat to the hot side of the generator was taking place at a rate of 5.5 watts. Determine the rate of energy transfer as heat from the cold side and the energy conversion efficiency.

Ans. $\dot{Q}_2 = 5.1$ watts, $\eta = 0.073$

- 5.6 A turbocompressor delivers $2.33 \text{ m}^3/\text{s}$ of air at 0.276 MPa, 43°C which is heated at this pressure to 430°C and finally expanded in a turbine which delivers 1860 kW. During the expansion, there is a heat transfer of 0.09 MJ/s to the surroundings. Calculate the turbine exhaust temperature if changes in kinetic and potential energy are negligible. Take $c_p = 1.005 \text{ kJ/kgK}$

Ans. 157°C

- 5.7 A reciprocating air compressor takes in $2 \text{ m}^3/\text{min}$ at 0.11 MPa, 20°C which it delivers at 1.5 MPa, 111°C to an aftercooler where the air is cooled at constant pressure to 25°C . The power absorbed by the compressor is 4.15 kW. Determine the heat transfer in (a) the compressor, and (b) the cooler. State your assumptions.

Ans. -0.17 kJ/s , -3.76 kJ/s

- 5.8 In a water cooling tower air enters at a height of 1 m above the ground level and leaves at a height of 7 m. The inlet and outlet velocities are 20 m/s and 30 m/s respectively. Water enters at a height of 8 m and leaves at a height of 0.8 m. The velocity of water at entry and exit are 3 m/s and 1 m/s respectively. Water temperatures are 80°C and 50°C at the entry and exit respectively. Air temperatures are 30°C and 70°C at the entry and exit respectively. The cooling tower is well insulated and a fan of 2.25 kW drives the air through the cooler. Find the amount of air per second required for 1 kg/s of water flow. The values of c_p of air and water are 1.005 and 4.187 kJ/kg K respectively.

Ans. 3.16 kg

- 5.9 Air at 101.325 kPa, 20°C is taken into a gas turbine power plant at a velocity of 140 m/s through an opening of 0.15 m^2 cross-sectional area. The air is compressed heated, expanded through a turbine, and exhausted at 0.18 MPa, 150°C through an opening of 0.10 m^2 cross-sectional area. The power output is 375 kW. Calculate the net amount of heat added to the air in kJ/kg. Assume that air obeys the law $p v = 0.287 (t + 273)$ where p is the pressure in kPa v is the specific volume in m^3/kg , and t is the temperature in $^\circ\text{C}$. Take $c_p = 1.005 \text{ kJ/kg K}$.

Ans. 150.23 kJ/kg

- 5.10 A gas flows steadily through a rotary compressor. The gas enters the compressor at a temperature of 16°C , a pressure of 100 kPa, and an enthalpy of 391.2 kJ/kg. The gas leaves the compressor at a temperature of 245°C , a pressure of 0.6 MPa, and an enthalpy of 534.5 kJ/kg. There is no heat transfer to or from the gas as it flows through the compressor. (a) Evaluate the external work done per unit mass of gas assuming the gas velocities at entry and exit to be negligible. (b) Evaluate the external work done per unit mass of gas when the gas velocity at entry is 80 m/s and that at exit is 160 m/s.

Ans. 143.3 kJ/kg, 152.9 kJ/kg

- 5.11 The steam supply to an engine comprises two streams which mix before entering the engine. One stream is supplied at the rate of 0.01 kg/s with an enthalpy of 2952 kJ/kg and a velocity of 20 m/s. The other stream is supplied at the rate of 0.1 kg/s with an enthalpy of 2569 kJ/kg and a velocity of 120 m/s. At the exit

from the engine the fluid leaves as two streams, one of water at the rate of 0.001 kg/s with an enthalpy of 420 kJ/kg and the other of steam; the fluid velocities at the exit are negligible. The engine develops a shaft power of 25 kW. The heat transfer is negligible. Evaluate the enthalpy of the second exit stream.

Ans. 2401 kJ/kg

- 5.12 The stream of air and gasoline vapour, in the ratio of 14:1 by mass, enters a gasoline engine at a temperature of 30°C and leaves as combustion products at a temperature of 790°C. The engine has a specific fuel consumption of 0.3 kg/kWh. The net heat transfer rate from the fuel-air stream to the jacket cooling water and to the surroundings is 35 kW. The shaft power delivered by the engine is 26 kW. Compute the increase in the specific enthalpy of the fuel-air stream, assuming the changes in kinetic energy and in elevation to be negligible.

Ans. - 1877 kJ/kg mixture

- 5.13 An air turbine forms part of an aircraft refrigerating plant. Air at a pressure of 295 kPa and a temperature of 58°C flows steadily into the turbine with a velocity of 45 m/s. The air leaves the turbine at a pressure of 115 kPa, a temperature of 2°C, and a velocity of 150 m/s. The shaft work delivered by the turbine is 54 kJ/kg of air. Neglecting changes in elevation, determine the magnitude and sign of the heat transfer per unit mass of air flowing. For air, take $c_p = 1.005$ kJ/kg K and the enthalpy $h = c_p t$.

Ans. + 7.96 kJ/kg

- 5.14 In a turbomachine handling an incompressible fluid with a density of 1000 kg/m³ the conditions of the fluid at the rotor entry and exit are as given below

	Inlet	Exit
Pressure	1.15 MPa	0.05 MPa
Velocity	30 m/s	15.5 m/s
Height above datum	10 m	2 m

If the volume flow rate of the fluid is 40 m³/s, estimate the net energy transfer from the fluid as work.

Ans. 60.3 MW

- 5.15 A room for four persons has two fans, each consuming 0.18 kW power, and three 100 W lamps. Ventilation air at the rate of 80 kg/h enters with an enthalpy of 84 kJ/kg and leaves with an enthalpy of 59 kJ/kg. If each person puts out heat at the rate of 630 kJ/h determine the rate at which heat is to be removed by a room cooler, so that a steady state is maintained in the room.

Ans. 1.92 kW

- 5.16 Air flows steadily at the rate of 0.4 kg/s through an air compressor, entering at 6 m/s with a pressure of 1 bar and a specific volume of 0.85 m³/kg, and leaving at 4.5 m/s with a pressure of 6.9 bar and a specific volume of 0.16 m³/kg. The internal energy of the air leaving is 88 kJ/kg greater than that of the air entering. Cooling water in a jacket, surrounding the cylinder absorbs heat from the air at the rate of 59 W. Calculate the power required to drive the compressor and the inlet and outlet cross-sectional areas.

(Ans. 45.4 kW, 0.057 m², 0.0142 m²)

- 5.17 Steam flowing in a pipeline is at a steady state represented by p_p, t_p, u_p, v_p, h_p and V_p . A small amount of the total flow is led through a small tube to an evacuated chamber which is allowed to fill slowly until the pressure is equal to the pipeline

pressure. If there is no heat transfer, derive an expression for the final specific internal energy in the chamber, in terms of the properties in the pipeline.

- 5.18 The internal energy of air is given, at ordinary temperatures, by

$$u = u_0 + 0.718 t$$

where u is in kJ/kg, u_0 is any arbitrary value of u at 0°C , kJ/kg, and t is temperature in $^\circ\text{C}$.

Also for air, $p v = 0.287 (t + 273)$

where p is in kPa and v is in m^3/kg .

- (a) An evacuated bottle is fitted with a valve through which air from the atmosphere, at 760 mm Hg and 25°C , is allowed to flow slowly to fill the bottle. If no heat is transferred to or from the air in the bottle, what will its temperature be when the pressure in the bottle reaches 760 mm Hg?

Ans. 144.2°C

- (b) If the bottle initially contains 0.03 m^3 of air at 400 mm Hg and 25°C , what will the temperature be when the pressure in the bottle reaches 760 mm Hg?

Ans. 71.6°C

- 5.19 A pressure cylinder of volume V contains air at pressure p_0 and temperature T_0 . It is to be filled from a compressed air line maintained at constant pressure p_1 and temperature T_1 . Show that the temperature of the air in the cylinder after it has been charged to the pressure of the line is given by

$$T = \frac{\gamma T_1}{1 + \frac{p_0}{p_1} \left(\gamma \frac{T_1}{T_0} - 1 \right)}$$

- 5.20 A small reciprocating vacuum pump having the rate of volume displacement V_d is used to evacuate a large vessel of volume V . The air in the vessel is maintained at a constant temperature T by energy transfer as heat. If the initial and final pressures are p_1 and p_2 respectively, find the time taken for the pressure drop and the necessary energy transfer as heat during evacuation. Assume that for air, $pV = mRT$, where m is the mass and R is a constant, and v is a function of T only.

$$\left[\text{Ans. } t = \frac{V}{V_d} \ln \frac{p_1}{p_2}; Q = (p_1 - p_2) V \right]$$

[Hint: $dm = -p(V_d dt)/(RT) = V dp/(RT)$].

- 5.21 A tank containing 45 kg of water initially at 45°C has one inlet and one exit with equal mass flow rates. Liquid water enters at 45°C and a mass flow rate of 270 kg/h. A cooling coil immersed in the water removes energy at a rate of 7.6 kW. The water is well mixed by a paddle wheel with a power input of 0.6 kW. The pressures at inlet and exit are equal. Ignoring changes in KE and PE, find the variation of water temperature with time.

Ans. $T = 318 - 22 [1 - \exp(-6t)]$

- 5.22 A rigid tank of volume 0.5 m^3 is initially evacuated. A tiny hole develops in the wall, and air from the surroundings at 1 bar, 21°C leaks in. Eventually, the pressure in the tank reaches 1 bar. The process occurs slowly enough that heat transfer between the tank and the surroundings keeps the temperature of the air inside the tank constant at 21°C . Determine the amount of heat transfer.

Ans. -50 kJ